

CHURCH LAW FORD, United Kingdom

WMO: 035440

Lat: 52.3589N

Lon: 1.3314W

Elev: 348

StdP: 14.51

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.8	(c) 27.3	(d) 22.2	(e) 16.9	(f) 29.8	(g) 24.4	(h) 18.7	(i) 31.1	(j) 25.4	(k) 46.4	(l) 22.7	(m) 46.0	(n) 3.6	(o) 30	(p) 0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	80.2	65.7	76.3	63.6	73.0	62.0	67.1	76.2	65.1	73.3	63.5	70.6	7.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.8	89.8	69.8	62.0	84.2	68.0	60.4	79.5	66.4	31.7	76.8	30.2	73.4	29.0	70.8	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	17.5	15.5	19.0	85.8	4.3	4.4	15.9	89.0	13.3	91.6	10.9	94.0	7.8	97.2
			18.8	71.9	4.2	4.8	15.8	75.4	13.3	78.1	11.0	80.8	7.9	84.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.1	39.7	40.2	43.4	47.6	52.8	58.4	62.4	62.1	57.4	51.5	44.7	40.3		
	DBStd	9.77	5.90	5.77	5.39	5.27	5.25	4.70	4.58	4.32	4.33	5.12	5.76	6.71		
	HDD50	1466	319	276	213	105	30	1	0	0	1	42	174	305		
	HDD65	5498	783	694	671	522	377	205	109	113	231	418	610	765		
	CDD50	1504	1	2	7	33	118	253	384	374	224	89	14	5		
	CDD65	62	0	0	0	0	0	7	28	23	4	0	0	0		
	CDH74	580	0	0	0	2	11	90	275	174	25	3	0	0		
	CDH80	104	0	0	0	0	0	13	56	34	1	0	0	0		
(14) Wind	WSAvg	7.1	8.1	8.3	8.4	7.6	7.2	6.3	6.0	6.0	6.1	6.8	7.0	7.4		
(15) Precipitation	PrecAvg	26.6	2.2	1.8	1.6	1.9	2.3	2.4	2.4	2.7	1.9	2.6	2.5	2.2		
	PrecMax	33.8	5.5	3.0	3.1	4.6	4.1	5.6	5.2	5.5	4.7	4.5	3.9	4.1		
	PrecMin	16.3	0.6	0.4	0.4	0.1	0.7	0.7	0.6	0.6	0.4	0.7	1.3	0.8		
	PrecStd	4.6	1.2	0.7	0.7	1.3	0.9	1.4	1.2	1.2	1.1	1.1	0.8	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.8	56.2	61.4	70.2	75.4	81.7	85.4	84.3	77.8	68.0	59.3	56.3		
		MCWB	51.5	52.1	51.8	57.1	62.5	66.6	67.5	68.7	64.2	60.9	56.3	53.4		
	2%	DB	52.7	52.9	57.2	64.1	71.0	76.2	80.8	79.1	72.1	63.7	57.3	54.4		
		MCWB	49.9	49.5	50.2	54.1	60.3	63.4	65.3	66.3	62.2	58.5	54.6	52.1		
	5%	DB	50.6	50.7	54.5	60.3	67.1	72.1	76.7	74.5	68.6	61.3	55.3	52.4		
		MCWB	48.1	47.7	49.0	52.2	58.2	61.5	63.5	63.2	59.8	57.1	52.6	50.3		
	10%	DB	48.6	48.9	52.2	57.2	62.9	68.5	73.0	71.0	65.7	59.4	53.2	50.3		
		MCWB	46.2	46.2	47.6	50.5	55.4	59.8	62.2	61.3	58.4	55.7	50.8	48.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.3	52.7	53.3	58.7	64.9	69.5	69.7	75.8	66.2	62.0	57.1	53.9		
		MCDB	54.2	55.1	58.2	67.3	71.6	77.7	81.4	76.4	73.3	66.8	58.6	55.4		
	2%	WB	50.4	50.1	51.6	55.7	61.6	65.0	66.8	66.7	63.5	59.6	55.1	52.5		
		MCDB	52.3	52.4	55.6	61.9	69.0	72.4	76.9	75.8	69.7	62.6	56.8	54.1		
	5%	WB	48.4	48.3	49.9	53.3	59.1	62.7	65.0	64.6	61.6	57.9	53.1	50.6		
		MCDB	50.3	50.3	53.5	58.6	65.2	69.7	74.0	72.0	66.7	60.6	54.9	52.3		
	10%	WB	46.5	46.5	48.2	51.2	56.8	60.8	63.2	62.8	59.7	56.1	51.1	48.3		
		MCDB	48.3	48.6	51.7	56.5	62.1	67.2	71.2	68.9	64.2	58.8	52.9	50.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.3	13.0	15.6	16.2	16.6	17.6	16.5	15.1	12.1	10.0	9.0		
		MCDBR	11.0	11.4	16.2	20.8	22.6	23.0	24.1	22.0	19.5	13.3	10.4	10.1		
	5% WB	MCWBR	10.0	9.2	10.6	12.4	13.3	12.5	11.0	11.0	10.7	9.0	9.0	9.3		
		MCWBR	10.7	11.2	14.3	18.9	19.8	19.9	20.9	18.9	16.7	12.2	10.2	10.0		
(40) Clear-Sky Solar Irradiance	taub		0.322	0.337	0.370	0.396	0.401	0.400	0.398	0.386	0.368	0.350	0.324	0.311		
	taud		2.356	2.381	2.287	2.225	2.250	2.279	2.289	2.349	2.400	2.430	2.407	2.336		
	Ebn at Noon		212	246	258	265	270	271	269	266	257	237	210	193		
	Edh at Noon		19	25	33	40	41	40	39	35	30	23	18	17		
(44) All-Sky Solar Radiation	RadAvg		243	462	801	1229	1513	1626	1561	1289	967	563	298	197		
	RadStd		21	56	98	135	97	160	167	93	86	49	33	21		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (100 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page